



Liar's Dominating Set in Unit Disk Graphs

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Abstract. In this article, we study a variant of the dominating set problem known as the liar's dominating set problem on unit disk graphs. We prove that the liar's dominating set problem is NP-complete and admits a polynomial time approximation scheme in unit disk graphs.

Keywords: Dominating set · Liar's dominating set · Unit disk graph
Approximation scheme

1 Introduction

Given a simple undirected graph $G = (V, E)$, the open and closed neighborhoods of a vertex v_i are defined by $N_G(v_i) = \{v_j \in V \mid (v_i, v_j) \in E \text{ and } v_i \neq v_j\}$ and $N_G[v_i] = N_G(v_i) \cup \{v_i\}$, respectively. A *dominating set* D of G is a subset of V such that every vertex in $V \setminus D$ is adjacent to at least one vertex in D . That is, each vertex $v_i \in V$ is either in D or there exists a vertex $v_j \in D$ such that $(v_i, v_j) \in E$. Observe that for any dominating set $D \subseteq V$, $|N_G[v_i] \cap D| \geq 1$ for each $v_i \in V$. We say that a vertex v_i is dominated by v_j in G , if $v_j \in D$ and $(v_i, v_j) \in E$. The dominating set problem asks to find a dominating set of minimum size in a given graph. A set $D \subseteq V$ is a *k-tuple dominating set* in G , if each vertex $v_i \in V$ is dominated by at least k vertices in D . In other words, $|N_G[v_i] \cap D| \geq k$ for each $v_i \in V$. The minimum cardinality of a k -tuple dominating set of a graph G is called the *k-tuple domination number* of G .

A *liar's dominating set* (LDS) of a simple undirected graph, $G = (V, E)$, is a dominating set D having the following two properties: (i) for every $v_i \in V$, $|N_G[v_i] \cap D| \geq 2$, and (ii) for every pair of distinct vertices v_i and v_j in V , $|(N_G[v_i] \cup N_G[v_j]) \cap D| \geq 3$. For a given graph G , the problem of finding an LDS in G of minimum cardinality is known as the *minimum liar's dominating set* (MLDS) problem. The cardinality of an MLDS in a graph G is known as the liar's domination number of G . Every 3-tuple dominating set is a liar's dominating set as it satisfies both conditions, so the liar's domination number lies between 2-tuple and 3-tuple domination numbers.

2 Related Work

The MLDS problem is introduced by Slater [14]. He showed that the problem is NP-hard for general graphs, and gave a lower bound on the liar's domination number in case of trees by proving that the size of any liar's dominating set of a tree of order n is between $\frac{3}{4}(n+1)$ and n . Later, Roden and Slater [13] characterized tree classes with liar's domination number equal to $\frac{3}{4}(n+1)$. In the same paper, they also showed that the MLDS problem is NP-hard even for bipartite graphs. Panda and Paul [9] proved that the problem is NP-hard for split graphs and chordal graphs. They also proposed a linear time algorithm for computing an MLDS in case of trees.

Panda et al. [12] studied the approximability of the problem and presented an $O(\ln \Delta(G))$ -factor approximation algorithm, where $\Delta(G)$ is the degree of the graph. Panda and Paul [10] considered the problem for proper interval graphs and proposed a linear time algorithm for computing a minimum cardinality liar's dominating set. The problem is also studied for bounded degree graphs, and p -claw free graphs [12]. Sterling [15] considered the problem on two-dimensional grid graphs and presented bounds on the liar's domination number.

Alimadadi et al. [1] provided the characterization of graphs and trees for which the liar's domination number is $|V|$ and $|V| - 1$, respectively. Panda and Paul [8, 11] studied variants of liar's domination, namely, connected liar's domination and total liar's domination. A *connected liar's dominating set* (CLDS) is an LDS whose induced subgraph is connected. A *total liar's dominating set* (TLDS) is a dominating set L with the following two properties: (i) for every $v \in V$, $|N_G(v) \cap L| \geq 2$, and (ii) for every distinct pair of vertices u and v , $|(N_G(u) \cup N_G(v)) \cap L| \geq 3$, where $N_G(\cdot)$ is the open neighborhood of a vertex. The objective of both problems is to find CLDS and TLDS of minimum size, respectively. They proved that both problems are NP-hard and proposed $O(\ln \Delta(G))$ -factor approximation algorithms. They also proved that the problems are APX-complete for graphs with maximum degree 4. Jallu and Das [6] studied the geometric version of the MLDS problem, and presented constant factor approximation algorithms.

2.1 Our Contribution

We study the MLDS problem on a geometric intersection graph model, particularly on unit disk graphs. A *unit disk graph* (UDG) is an intersection graph of disks of equal radii in the plane. Given a set $S = \{d_1, d_2, \dots, d_n\}$ of n circular disks in the plane, each having diameter 1, the corresponding UDG $G = (V, E)$ is defined as follows: each vertex $v_i \in V$ corresponds to a disk $d_i \in S$, and there is an edge between two vertices if and only if the Euclidean distance between the respective disk centers is at most 1.

Our interest in this problem arises from the following scenario. Consider a graph in which each node is a possible location for an intruder such as a thief, or a saboteur. We would like to detect and report the intruder's location in the graph. A protection device such as a camera, or a sensor placed at a node can not

only detect (and report) the intruder's presence at it, but also at its neighbors. Our objective is to place a minimum number of protection devices such that the intrusion of the intruder at any vertex is detected and reported. In this situation, one must place the devices at the vertices of a minimum dominating set of the graph to achieve the goal. The protection devices are prone to failure and hence certain degree of redundancy is needed in the solution. Also, some times the devices may misreport the intruder's location deliberately or due to transmission error. Assume that at most one protection device in the closed neighborhood of the intruder can lie (misreport). In this context, one must place the protection devices at the vertices of an MLDS of the graph to achieve the objective. The first property in the definition of LDS deals with single device fault-tolerance, while the second property deals with the case in which two distinct locations about the intruder are reported.

3 Hardness of the MLDS Problem on UDGs

In this section, we show that the MLDS problem on UDGs is NP-complete by reducing the *vertex cover* problem defined on planar graphs to it, which is known to be NP-complete [3]. The decision versions of both problems are defined below.

The MLDS problem on UDGs (LDS-UDG)

Instance: A unit disk graph $G = (V, E)$ and a positive integer k .

Question: Does there exist a liar's dominating set L in G such that $|L| \leq k$?

The vertex cover problem on planar graphs (VC-PLA)

Instance: An undirected planar graph G with maximum degree 3 and a positive integer k .

Question: Does there exist a vertex cover D of G such that $|D| \leq k$?

Lemma 1 ([16]). *A planar graph $G = (V, E)$ with maximum degree 4 can be embedded in the plane using $O(|V|^2)$ area in such a way that its vertices are at integer coordinates and its edges are drawn so that they are made up of line segments of the form $x = i$ or $y = j$, for integers i and j .*

This kind of embedding is known as *orthogonal drawing* of a graph. Biedl and Kant [2] gave a linear time algorithm that produces an orthogonal drawing of a given graph with the property that the number of bends along each edge is at most 2 (see Fig. 1).

Corollary 1. *A planar graph $G = (V, E)$ with maximum degree 3 and $|E| \geq 2$ can be embedded in the plane such that its vertices are at $(4i, 4j)$ and its edges are drawn as a sequence of consecutive line segments on the lines $x = 4i$ or $y = 4j$, for integers i and j .*

Lemma 2. *Let $G = (V, E)$ be an instance of VC-PLA with $|E| \geq 2$. An instance $G' = (V', E')$ of LDS-UDG can be constructed from G in polynomial-time.*

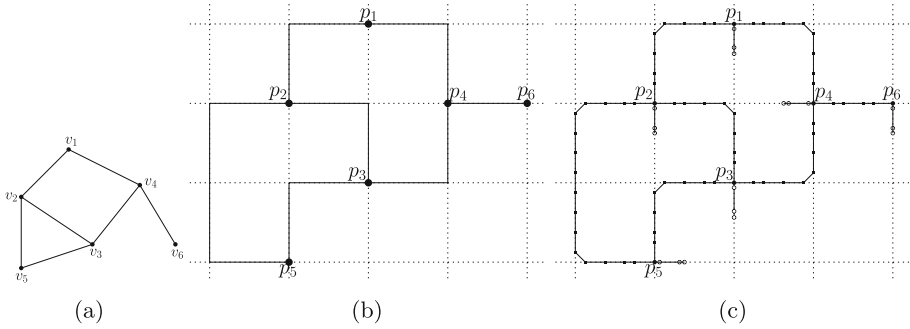


Fig. 1. (a) A planar graph G with maximum degree 3, (b) its embedding on a grid, and (c) a UDG construction from the embedding.

Proof. We construct G' in four phases.

Phase 1: Embedding of G into a grid of size 4×4

Embed the instance G in the plane as discussed previously using one of the algorithms in [4, 5]. An edge in the embedding is a sequence of connected line segment(s) of length four units each. If the total number of line segments used in the embedding is ℓ , then the sum of the lengths of the line segments is 4ℓ as each line segment has length 4 units. We name the points in the embedding correspond to the vertices of G by *node points*.

Phase 2: Adding extra points to the embedding

Divide the set of line segments in the embedding into two categories, namely, proper and improper. We call a line segment *proper* if none of its end points correspond to a vertex in G . A line segment is *improper* if it is not a proper segment. For each edge (p_i, p_j) of length 4 units we add two points at distances 1 and 1.5 units of p_i and p_j , respectively (thus adding four points in total, see the edge (p_4, p_6) in Fig. 1(c)). For each edge of length greater than 4 units, we also add points as follows: for each improper line segment we add four points at distances 1, 1.5, 2.5, and 3.5 units from the end point corresponding to a vertex in G , and for each proper line segment we add four points at distances 0.5 and 1.5 units from its end points (see Fig. 1(c)). We name the points added in this phase *joint points*.

Phase 3: Adding extra line segments and points

Add a line segment of length 1.4 units (on the lines $x = 4i$ or $y = 4j$ for some integers i or j) for every point p_i , which corresponds to a vertex v_i in G , without coinciding with the line segments that had already been drawn. Observe that adding this line segment on the lines $x = 4i$ or $y = 4j$ is possible without losing the planarity as the maximum degree of G is 3. Now, add three points (say x_i , y_i , and z_i) on these line segments at distances 0.2, 1.2, and 1.4 units, respectively, from p_i . We name the points added in this phase *support points*.

Phase 4: Construction of UDG

For convenience, let us denote the set of node points, joint points, and support points by N , J , and S , respectively. Let $N = \{p_i \mid v_i \in V\}$, $J = \{q_1, q_2, \dots, q_{4\ell}\}$,

and $S = \{x_i, y_i, z_i \mid v_i \in V\}$. We construct a UDG $G' = (V', E')$, where $V' = N \cup J \cup S$ and there is an edge between two points in V' if and only if the Euclidean distance between the points is at most 1 (see Fig. 1(c)). Observe that, $|N| = |V| (= n)$, $|J| = 4\ell$, where ℓ is the total number of line segments in the embedding, and $|S| = 3|V| (= 3n)$. Hence, $|V'| = 4(n + \ell)$ and ℓ is bounded by a polynomial of n . Therefore G' can be constructed in polynomial-time.

Theorem 1. *LDS-UDG is NP-complete.*

Proof. For any given set $L \subseteq V$ and a positive integer k , we can verify whether L is a liar's dominating set of size at most k or not in polynomial-time.

We prove the hardness of LDS-UDG by reducing VC-PLA to it. Let $G = (V, E)$ be an instance of VC-PLA. Construct an instance $G' = (V', E')$ of LDS-UDG as discussed in Lemma 2. We now prove the following claim: *G has a vertex cover of size at most k if and only if G' has a liar's dominating set of size at most $k + 3\ell + 3n$.*

Necessity: Let $D \subseteq V$ be a vertex cover of G such that $|D| \leq k$. Let $N' = \{p_i \in N \mid v_i \in D\}$, i.e., N' is the set of vertices in G' that correspond to the vertices in D . From each segment in the embedding we choose 3 vertices. The set of chosen vertices, say $J' (\subseteq J)$, together with N' and S will form an LDS of desired cardinality in G' . We now discuss the process of obtaining the set J' . Initially $J' = \emptyset$. As D is a vertex cover, every edge in G has at least one of its end vertices in D . Let (v_i, v_j) be an edge in G and $v_i \in D$ (the tie can be broken arbitrarily if both v_i and v_j are in D). Note that the edge (v_i, v_j) is represented as a sequence of line segments in the embedding. Start traversing the segments (of (v_i, v_j)) from p_i , where p_i corresponds to v_i , and add all the vertices to J' except the first one from each segment encountered in the traversal (see (p_2, p_5) in Fig. 2b. The big vertices are part of J' while traversing from p_2).

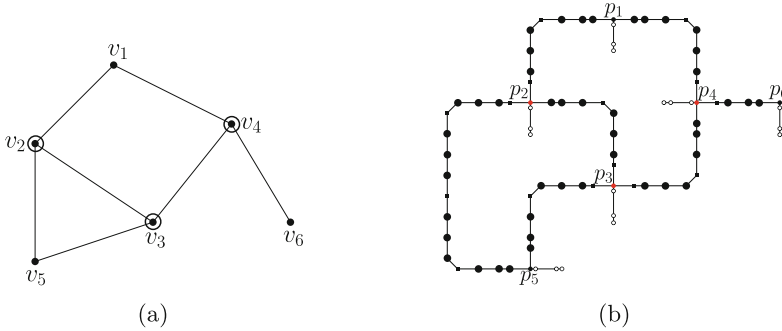


Fig. 2. (a) A vertex cover $\{v_2, v_3, v_4\}$ in G , and (b) the construction of J' in G' (the tie between v_2 and v_3 , and v_3 and v_4 is broken by choosing v_3)

Apply the above process to each edge in G . Observe that the cardinality of J' is 3ℓ as we have chosen 3 vertices from each segment in the embedding. Let $L = N' \cup J' \cup S$. Now, we argue that L is a liar's dominating set in G' .

1. Each $p_i \in N$ is dominated by x_i in S . If $p_i \in N'$ (i.e., the corresponding vertex $v_i \in D$ in G), then $|N_{G'}[p_i] \cap L| \geq |\{p_i, x_i\}| = 2$. If $p_i \notin N'$, then there must exist at least one vertex q_j in J' dominating p_i . The existence of q_j is guaranteed by the way we constructed J' . Hence, $|N_{G'}[p_i] \cap L| \geq |\{q_j, x_i\}| = 2$. In either case every vertex in N is dominated by at least two vertices in L . It is needless to say that vertex in J is dominated by at least two vertices in $N' \cup J'$. Similarly, every vertex in S is dominated by itself, by its neighbor(s) in S , and, perhaps, by one vertex in N' . Therefore, every vertex in V' is double dominated by vertices in L .
2. Consider a pair of distinct vertices in V' . Of course, every pair of distinct vertices in S satisfy the liar's second condition. We prove that remaining pairs of distinct vertices also satisfy the liar's second condition by considering all possible cases.

Case a. $p_i, p_j \in N$: If at least one of p_i, p_j belongs to N' (without loss of generality say $p_i \in N'$), then $|(N_{G'}[p_i] \cup N_{G'}[p_j]) \cap L| \geq |\{x_i, x_j, p_i\}| = 3$. If none of p_i, p_j belongs to N' , then there must exist some $q_i, q_j \in J'$ such that q_i, q_j dominate p_i, p_j , respectively. Hence, $|(N_{G'}[p_i] \cup N_{G'}[p_j]) \cap L| \geq |\{x_i, x_j, q_i, q_j\}| = 4$.

Case b. $q_i, q_j \in J$: If both $q_i, q_j \in J'$, then it is trivial that $|(N_{G'}[q_i] \cup N_{G'}[q_j]) \cap L| \geq 3$. Suppose one of q_i, q_j belongs to J' (without loss of generality say $q_i \in J'$). As every vertex in G' is double dominated, q_j must be dominated by two vertices in J' or by either some q_k in J' and some p_l in N' . In either case we get $|(N_{G'}[q_i] \cup N_{G'}[q_j]) \cap L| \geq 3$. A similar argument works even if none of q_i, q_j belong to J' .

Case c. $p_i \in N$ and $q_j \in J$: If none of p_i and q_j belong to L , then the argument is trivial as each one is dominated by at least two vertices in L . If both belong to L , then $|(N_{G'}[p_i] \cup N_{G'}[q_j]) \cap L| \geq |\{p_i, x_i, q_j\}| = 3$. If $p_i \in L$ and $q_j \notin L$ (the other case is similar), then $|(N_{G'}[p_i] \cup N_{G'}[q_j]) \cap L| \geq 3$ holds as q_j is double dominated.

Likewise, we can argue for other pair combinations too. Therefore, every pair of distinct vertices in V' is dominated by at least 3 vertices in L .

Thus, we have L is an LDS in G' and $|L| = |N'| + |J'| + |S| \leq k + 3\ell + 3n$.

Sufficiency: Let $L \subseteq V'$ be an LDS of size at most $k + 3\ell + 3n$. We prove that G has a vertex cover of size at most k with the aid of the following claims: (i) $S \subset L$, (ii) Every segment in the embedding must contribute at least 3 vertices to L and hence $|J \cap L| \geq 3\ell$, where ℓ is the total number of segments in the embedding, and (iii) If p_i and p_j correspond to end vertices of an edge (v_i, v_j) in G , and both p_i, p_j are not in L , then there must be at least $3\ell' + 1$ vertices in L form the segment(s) representing the edge (v_i, v_j) , where ℓ' is the number of segments representing the edge (v_i, v_j) in the embedding.

We shall show that, by removing and/or replacing some vertices in L , a set of k vertices from N can be chosen such that the corresponding vertices in G is a vertex cover. The vertices in S account for $3n$ vertices in L (due to Claim (i)). Let $L = L \setminus S$ and $D = \{v_i \in V \mid p_i \in L \cap N\}$. If any edge (v_i, v_j) in G has none of its end vertices in D , then we do the following: consider the sequence of segments representing the edge (v_i, v_j) in the embedding. Since, both p_i and p_j are not in L , there must exist a segment having all its vertices in L (due to Claim (iii)). Consider the segment having its four vertices in L . Delete any one of the vertices on the segment and introduce p_i (or p_j). Update D and repeat the process till every edge has at least one of its end vertices in D . (due to Claim (ii)) D is a vertex cover in G with $|D| \leq k$. Therefore, LDS-UDG is NP-complete.

4 Approximation Scheme

We propose a PTAS for the MLDS problem on UDGs, i.e., for a given UDG $G = (V, E)$ and a parameter $\epsilon > 0$, we propose an algorithm which produces a liar's dominating set of size no more than $(1 + \epsilon)$ times the size of a minimum liar's dominating set in G . We use $\delta_G(u, v)$ to denote the number of edges on a shortest path between u and v in G . For $A, B \subseteq V$, $\delta_G(A, B)$ denotes the distance between A and B and is defined as $\delta_G(A, B) = \min_{u \in A, v \in B} \{\delta_G(u, v)\}$. For $A \subseteq V$, $LD(A)$ and $LD_{opt}(A)$ denote an LDS and an optimal (minimum size) LDS of A in G , respectively. We define the closed neighborhood of a set $A \subseteq V$ as $N_G[A] = \bigcup_{v \in A} N_G[v]$.

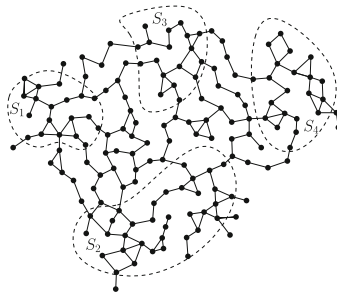


Fig. 3. A 4-separated collection $S = \{S_1, S_2, S_3, S_4\}$

The proposed PTAS is based on the concept of m -separated collection of subsets of V ($m \geq 4$). Let $G = (V, E)$ be a UDG. A collection $S = \{S_1, S_2, \dots, S_k\}$ such that $S_i \subseteq V$ for $i = 1, 2, \dots, k$, is said to be an m -separated collection, if $\delta_G(S_i, S_j) > m$, for $1 \leq i \leq k$ and $1 \leq j \leq k$ (see Fig. 3 for a 4-separated collection). Nieberg and Hurink [7] considered 2-separated collection to propose a PTAS for the minimum dominating set problem on unit disk graphs.

Lemma 3. *Let $S = \{S_1, S_2, \dots, S_k\}$ be an m -separated collection. If $|S_i| \geq 3$ for $1 \leq i \leq k$, then $\sum_{i=1}^k |LD_{opt}(S_i)| \leq |LD_{opt}(V)|$.*

Proof. Observe that $N_G[S_i] \cap N_G[S_j] = \emptyset$ for $i \neq j$ and $1 \leq i, j \leq k$. Also, $LD_{opt}(S_i) \cap LD_{opt}(S_j) = \emptyset$ as S_i and S_j are m -separated. Let $S'_i = \{u \in V \mid v \in S_i \text{ and } \delta_G(u, v) \leq 2\}$ for $i = 1, 2, \dots, k$. Observe that $S_i \subseteq S'_i$ and $S'_i \cap LD_{opt}(V)$ is a liar's dominating set of S_i for $i = 1, 2, \dots, k$. Since, $\delta_G(S_i, S_j) > m (\geq 4)$ for $i \neq j$, implies $S'_i \cap S'_j = \emptyset$. Therefore, $(S'_i \cap LD_{opt}(V)) \cap (S'_j \cap LD_{opt}(V)) = \emptyset$ and $\bigcup_{i=1}^k (S'_i \cap LD_{opt}(V)) \subseteq LD_{opt}(V)$. Also, $LD_{opt}(S_i) \subseteq S'_i \cap LD_{opt}(V)$ for $i = 1, 2, \dots, k$, $S'_i \cap LD_{opt}(V)$ is a liar's dominating set of S_i , and $LD_{opt}(S_i)$ is a minimum size liar's dominating set. Thus, $\bigcup_{i=1}^k LD_{opt}(S_i) \subseteq \bigcup_{i=1}^k (S'_i \cap LD_{opt}(V)) \subseteq LD_{opt}(V)$. Hence, the result of the lemma follows.

Lemma 4. *Let $S = \{S_1, S_2, \dots, S_k\}$ be an m -separated collection, and $N_1, N_2, \dots, N_k$ be subsets of V with $S_i \subseteq N_i$ for all $i = 1, 2, \dots, k$. If there exists $\rho \geq 1$ such that $|LD_{opt}(N_i)| \leq \rho |LD_{opt}(S_i)|$ holds for all $i = 1, 2, \dots, k$, and if $\bigcup_{i=1}^k LD_{opt}(N_i)$ is a liar's dominating set in G , then the value of $\sum_{i=1}^k |LD_{opt}(N_i)|$ is at most ρ times the size of a minimum liar's dominating set in G .*

In the proposed PTAS we have chosen the concept of 4-separated collection to make $LD_{opt}(N_i) \cap LD_{opt}(N_j) = \emptyset$ for $i \neq j$.

4.1 Algorithm

In this section, we discuss the construction of a 4-separated collection $S = \{S_1, S_2, \dots, S_k\}$ and subsets $N_1, N_2, \dots, N_k$ of V such that $S_i \subseteq N_i$ for all $i = 1, 2, \dots, k$. The algorithm proceeds in an iterative manner. Initially $V_1 = V$. In the i -th iteration the algorithm computes S_i and N_i . For a given $\epsilon > 0$, the i -th iteration of the algorithm starts with an arbitrary vertex $v \in V_i$ and increases the value of $r (= 1, 2, \dots)$ as long as $|LD(N_G^{r+4}[v])| > \rho |LD(N_G^r[v])|$ holds. Here, $LD(N_G^{r+4}[v])$ and $LD(N_G^r[v])$ are liar's dominating sets of $N_G^{r+4}[v]$ and $N_G^r[v]$, respectively, and $\rho = 1 + \epsilon$. The smallest r violating the above condition, say $\hat{r}$, is obtained. Set $S_i = N_G^{\hat{r}}[v]$ and $U_i = N_G^{\hat{r}+4}[v]$. Now, the removal of U_i from V_i may lead to some isolated (i) vertex $u \in V_i$, and/or (ii) connected component with two vertices $u, w \in V_i$. In case (i), for each such vertex u find $x, y \in U_i$ such that $\{u, x, y\}$ forms a connected component and update U_i as follows: $U_i = U_i \setminus \{x, y\}$. In case (ii), for each such pair of vertices u, w find $x \in U_i$ such that $\{u, w, x\}$ forms a connected component and update U_i as follows: $U_i = U_i \setminus \{x\}$. Set $N_i = U_i$ and $V_{i+1} = V_i \setminus N_i$. The process stops if $V_{i+1} = \emptyset$ and returns the sets S_i s and N_i s. The collection of the sets S_i s is a 4-separated collection. The pseudo code is given in Algorithm 1.

The liar's dominating set of a r -th neighborhood of a vertex v , $LD(N_G^r[v])$, can be computed with respect to G as follows. We successively find maximal independent sets I_1, I_2 and I_3 such that $I_1 \cap I_2 \cap I_3 = \emptyset$. Now $I_1 \cup I_2 \cup I_3$ is a liar's dominating set for $N_G^r[v] \setminus I_1$ as every vertex not in I_1 either belongs to $I_2 \cup I_3$ or is adjacent to at least one vertex in each I_1, I_2 , and I_3 . To ensure the liar's domination conditions for the vertices in I_1 , for each vertex u in I_1 we add two arbitrary vertices from the neighborhood of u , if they exist. If u

Algorithm 1. Liar's dominating set**Require:** An undirected graph $G = (V, E)$ with $|V| \geq 3$ and an arbitrary small $\epsilon > 0$ **Ensure:** A liar's dominating set $\mathcal{D}$ of V

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1:  $i \leftarrow 0$  and  $V_{i+1} \leftarrow V$ 
2:  $\mathcal{D} \leftarrow \emptyset$  and  $\rho \leftarrow 1 + \epsilon$ 
3: while ( $V_{i+1} \neq \emptyset$ ) do
4:   pick an arbitrary  $v \in V_{i+1}$ 
5:    $N^0[v] \leftarrow v$  and  $r \leftarrow 1$ 
6:   while  $|LD(N_G^{r+4}[v])| > \rho |LD(N_G^r[v])|$  do ▷ call Algorithm 2
7:      $r \leftarrow r + 1$ 
8:    $\hat{r} \leftarrow r$  ▷ the smallest  $r$  violating while condition in step 6
9:    $i \leftarrow i + 1$  ▷ the index  $i$  keeps track of the number of iterations
10:   $S_i \leftarrow N_G^{\hat{r}}[v]$  and  $U_i \leftarrow N_G^{\hat{r}+4}[v]$ 
11:  if ( $V_{i+1} \setminus N_G^{\hat{r}+4}[v]$  contains isolated components of size 1 and/or 2) then
12:    for (each component,  $\{u\}$ , of size 1) do
13:      find  $x, y \in U_i$  such that  $\{u, x, y\}$  is a connected component
14:       $U_i \leftarrow U_i \setminus \{x, y\}$ 
15:    for (each component,  $\{u, w\}$ , of size 2) do
16:      find  $x \in U_i$  such that  $\{u, w, x\}$  is a connected component
17:       $U_i \leftarrow U_i \setminus \{x\}$ 
18:   $N_i \leftarrow U_i$ 
19:   $\mathcal{D} \leftarrow \mathcal{D} \cup LD(N_i)$  ▷ call Algorithm 2
20:   $V_{i+1} \leftarrow V_i \setminus N_i$ 
21: return  $\mathcal{D}$ 

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has only one neighbor, say u' , then we add u' and one of its neighbors in the solution. The pseudo code is given in Algorithm 2. In summary, Algorithm 1 deals with obtaining an m -separated collection $S = \{S_1, S_2, \dots, S_k\}$ and collection $N = \{N_1, N_2, \dots, N_k\}$ such that $S_i \subseteq N_i \subseteq V$ and using Algorithm 2 (that deals with obtaining a liar's dominating set of the r -th neighborhood of a vertex) it computes a liar's dominating set for G .

Lemma 5. $LD(N_G^r[v])$ returned by Algorithm 2 is an LDS of $N_G^r[v]$ in G .

Proof. Let $u \in V$. We prove the first condition of liar's domination in the following cases.

Case 1. $u \in N_G^r[v] \setminus (I_1 \cup I_2 \cup I_3)$

Observe that $I_1 \cap I_2 \cap I_3 = \emptyset$. The vertex u is dominated by at least three vertices as in each round u is dominated by at least one vertex (see **for** loop in line number 2 in Algorithm 2). Thus, $|N_G[u] \cap LD(N_G^r[v])| \geq 3$.

Case 2. $u \in I_1 \cup I_2 \cup I_3$

We consider the following two sub-cases: (a) $u \in I_2 \cup I_3$: In this case $|N_G[u] \cap LD(N_G^r[v])| \geq 2$ holds as every vertex in $I_2 \cup I_3$ has at least one neighbor in I_1 . (b) $u \in I_1$: For every vertex u in I_1 , $LD(N_G^r[v])$ contains a neighbor of u or u 's neighbor's neighbor (see line number 6 in Algorithm 2). Hence, $|N_G[u] \cap LD(N_G^r[v])| \geq 2$ holds.

Algorithm 2. Liar's dominating set in r -th neighborhood**Require:** The r -th neighborhood of a vertex v i.e., $N_G^r[v]$ **Ensure:** A liar's dominating set $LD(N_G^r[v])$ of $N_G^r[v]$

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1:  $j \leftarrow 0, I_j \leftarrow \emptyset, LD(N_G^r[v]) \leftarrow \emptyset$  and  $N \leftarrow N_G^r[v]$ 
2: for ( $j = 1$  to  $3$ ) do
3:   if ( $N \neq \emptyset$ ) then
4:      $I_j \leftarrow MIS(N \setminus I_{j-1})$   $\triangleright MIS(\cdot)$  returns a maximal independent set
5:      $LD(N_G^r[v]) \leftarrow LD(N_G^r[v]) \cup I_j; N \leftarrow N \setminus I_j$ 
6: for every  $u \in I_1$  do
7:   if ( $|N_G[u]| > 2$ ) then
8:     add two arbitrary vertices from  $N_G[u]$  to  $LD(N_G^r[v])$ 
9:   else
10:    add  $w$  and its neighbor to  $LD(N_G^r[v])$   $\triangleright w$  is the neighbor of  $u$ 
11: return  $LD(N_G^r[v])$ 

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Now we prove the second condition of liar's domination. Let $u, w \in N_G^r[v]$. We prove the inequality $|(N_G[u] \cup N_G[w]) \cap LD(N_G^r[v])| \geq 3$ by considering the following cases.

Case 1. $u, w \in N_G^r[v] \setminus (I_1 \cup I_2 \cup I_3)$: Observe that both u and w have a neighbor in each I_1, I_2 and I_3 . Hence $|(N_G[u] \cup N_G[w]) \cap LD(N_G^r[v])| \geq 3$.

Case 2. $u \in N_G^r[v] \setminus (I_1 \cup I_2 \cup I_3)$ and $w \in I_1 \cup I_2 \cup I_3$ (the other case proof is similar): In this case, the inequality $|(N_G[u] \cup N_G[w]) \cap LD(N_G^r[v])| \geq 3$ is true as $|N_G[u] \cap LD(N_G^r[v])| \geq 3$ and $|N_G[w] \cap LD(N_G^r[v])| \geq 2$.

Case 3. $u, w \in I_1 \cup I_2 \cup I_3$: The inequality $|(N_G[u] \cup N_G[w]) \cap LD(N_G^r[v])| \geq 3$ holds in this case as $u, w \in LD(N_G^r[v])$ and each of them has a neighbor (different from w, u , respectively) in $LD(N_G^r[v])$. This completes the proof of the lemma.

Lemma 6. $|LD(N_G^r[v])| \leq O(r^2)$.

Proof. Algorithm 2 computes $LD(N_G^r[v])$ by first computing maximal independent sets I_1, I_2 , and I_3 subsequently. After computing I_1, I_2 , and I_3 , the algorithm adds two vertices for each vertex in I_1 to ensure that $LD(N_G^r[v])$ is a feasible solution. Without loss of generality we assume that $|I_1| \geq |I_2|, |I_3|$. Hence, $|LD(N_G^r[v])| = 3|I_1| + |I_2| + |I_3| \leq 5 \times |I_1| \leq 5 \times \frac{\pi(r+1)^2}{\pi(1)^2} = O(r^2)$. The latter inequality follows from the standard area argument, the number of non-intersecting unit disks can be packed in a larger disk of radius $r + 1$ centered at v .

Lemma 7. In each iteration of Algorithm 1, there exists an r violating the condition $|(LD(N_G^{r+4}[v]))| > \rho |LD(N_G^r[v])|$, where $\rho = 1 + \epsilon$.

Proof. We prove the lemma by contradiction. Suppose there exists $v \in V$ such that $|(LD(N_G^{r+4}[v]))| > \rho |LD(N_G^r[v])|$ for $r = 1, 2, \dots$. Observe that $|LD(N_G^2[v])| \geq 3$ as there exists at least three vertices in G .

If r is even,

$(r+5)^2 \geq |(LD(N_G^{r+4}[v])| > \rho |LD(N_G^r[v])| > \cdots > \rho^{\frac{r}{2}} |LD(N_G^2[v])| \geq 3\rho^{\frac{r}{2}}$, and if r is odd,
 $(r+5)^2 \geq |(LD(N_G^{r+4}[v])| > \rho |LD(N_G^r[v])| > \cdots > \rho^{\frac{r-1}{2}} |LD(N_G^3[v])| \geq 3\rho^{\frac{r-1}{2}}$.

In both the cases the first inequality follows from Lemma 6. Hence,

$$(r+5) > \begin{cases} (\sqrt{\rho})^r, & \text{if } r \text{ is even} \\ (\sqrt{\rho})^{r-1}, & \text{if } r \text{ is odd} \end{cases} \quad (1)$$

The right hand part in inequality (1) is an exponential function in r and the left hand part is a polynomial in r , for arbitrarily large r none of the inequalities can be true. Hence we arrived at contradiction. Thus there exists an r violating the condition.

Lemma 8. *The smallest r violating inequality (1) is a constant and is bounded by $O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$.*

Proof. Lemma 7 suggests that the smallest r violating the condition cannot be sufficiently large but a constant. Let $\hat{r}$ be the smallest r violating the condition i.e., when $r = \hat{r}$ the inequalities in (1) are violated. Using the inequality $\log(1 + \epsilon) > \frac{\epsilon}{2}$ for $0 < \epsilon < 1$, we have $\hat{r} \leq O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$ as follows.

Consider the inequality $(1 + \epsilon)^x < x^2$ for a fixed $\epsilon > 0$. We prove that there exist ϵ' such that $(1 + \epsilon)^x < x^2 < (x + 5)^2$ for $0 < \epsilon < \epsilon'$. The latter inequality is trivial. Let $x = \frac{4}{\epsilon} \log \frac{1}{\epsilon}$. By taking the logarithm on both sides of the former inequality, we get, $\log 4 + \log \log \frac{1}{\epsilon} > 0$. Note that, we can always find an ϵ' such that $\log 4 + \log \log \frac{1}{\epsilon} > 0$ for $0 < \epsilon < \epsilon'$. Therefore, $(1 + \epsilon)^x < x^2$ holds for sufficiently smaller ϵ values and hence, $\hat{r} \leq \frac{4}{\epsilon} \log \frac{1}{\epsilon}$.

Lemma 9. *For a given $v \in V$, liar's dominating set $LD_{opt}(N_i)$ of N_i can be computed in polynomial time.*

Proof. Note that $N_i \subseteq N_G^{r+4}[v]$. The size of a liar's dominating set $LD(N_i)$ of N_i obtained by Algorithm 2 is bounded by $O(r^2)$ (by Lemma 6). Again, $r = O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$ by Lemma 8. Therefore, the size of the minimum size liar's dominating set $LD_{opt}(N_i)$ of N_i is bounded by a constant. The process of checking whether a given set is a liar's dominating set or not can be done in polynomial-time. Therefore, we can consider every subset of N_i as a possible liar's dominating set and check whether it is a liar's dominating set or not in polynomial-time. Finally, the minimum size liar's dominating set is reported. Thus the lemma.

Lemma 10. *For the collection of neighborhoods $\{N_1, N_2, \dots, N_k\}$ created by Algorithm 1, the union $\mathcal{D} = \bigcup_{i=1}^k LD(N_i)$ is a liar's dominating set in G .*

Proof. We first prove that for every $v \in V$, $|N_G[v] \cap \mathcal{D}| \geq 2$. Observe that $\bigcup_{i=1}^k N_i = V$ as $V_{i+1} = V_i \setminus N_i$ and $N_i \subseteq V_i$. Thus, every vertex $v \in N_i$ for some $1 \leq i \leq k$. By Lemma 5, $|N_G[v] \cap \mathcal{D}| \geq 2$ is satisfied.

Now we prove the second condition. Consider two arbitrary vertices $u, v \in V$. The following cases may arise.

Case 1. $u, v \in N_i$ for some $1 \leq i \leq k$

Since $LD(N_i)$ is the liar's dominating set of N_i in G , we have, $|(N_G[u] \cup N_G[v]) \cap LD(N_i)| \geq 3$ for every $u, v \in N_i$. Hence, $|(N_G[u] \cup N_G[v]) \cap \mathcal{D}| \geq 3$ for every $u, v \in V$.

Case 2. $u \in N_i$ and $v \in N_j$ for some $i \neq j$ and $1 \leq i, j \leq k$

If u and v are not adjacent in G , the proof is trivial. Hence, we assume that $(u, v) \in E$ i.e., u and v are adjacent in G . Now the following sub-cases may arise.

(a) $u \in LD(N_i)$ and $v \in LD(N_j)$

Observe that $|N_G[u] \cap LD(N_i)| \geq 2$ and $|N_G[v] \cap LD(N_j)| \geq 2$ as $LD(N_i)$ and $LD(N_j)$ are liar's dominating sets of N_i and N_j , respectively. Hence, u has a neighbor, say w , in $LD(N_i)$, similarly v has also a neighbor, say x , in $LD(N_j)$. However, maybe $w = x$ or maybe not. In either case $|(N_G[u] \cup N_G[v]) \cap \mathcal{D}| \geq 3$ holds.

(b) $u \notin LD(N_i)$ and $v \in LD(N_j)$ (the other case proof similar)

Since $LD(N_i)$ is a liar's dominating set of N_i , we have $|N_G[u] \cap LD(N_i)| \geq 2$. Hence, $|(N_G[u] \cup N_G[v]) \cap \mathcal{D}| \geq 3$ is true as v is part of the solution.

(c) $u \notin LD(N_i)$ and $v \notin LD(N_j)$ (The proof is similar to the previous cases).

Corollary 2. For the collection $N = \{N_1, N_2, \dots, N_k\}$ created by Algorithm 1, the union $\mathcal{D}^* = \bigcup_{i=1}^k LD_{opt}(N_i)$ is a liar's dominating set.

Theorem 2. For a given UDG, $G = (V, E)$, and an $\epsilon > 0$, we can design a $(1 + \epsilon)$ -factor approximation algorithm to find an LDS in G with running time $n^{O(c^2)}$, where $c = O(\frac{1}{\epsilon} \log \frac{1}{\epsilon})$.

Proof. Note that Algorithm 1 generates the collection of sets $S = \{S_1, S_2, \dots, S_k\}$ and $N = \{N_1, N_2, \dots, N_k\}$ such that S is a 4-separated collection of V with $S_i \subseteq N_i$ for each $i \in \{1, 2, \dots, k\}$ and $\bigcup_{i=1}^k N_i = V$ with $N_i \cap N_j = \emptyset$ for $i \neq j$. Corollary 2 suggests that $\mathcal{D}^* = \bigcup_{i=1}^k LD_{opt}(N_i)$ is a liar's dominating set of G . The approximation bound follows from Lemmas 3 and 4. Let $|N_i| = n_i$ for $1 \leq i \leq k$. By Lemma 9, an optimal liar's dominating set $LD_{opt}(N_i)$ of N_i can be computed in $n_i^{O(\frac{1}{\epsilon^2} \log \frac{1}{\epsilon})}$ time. Therefore, the total running time to compute $\mathcal{D}^*$ is $\sum_{i=1}^k n_i^{O(\frac{1}{\epsilon^2} \log \frac{1}{\epsilon})} \leq n^{O(\frac{1}{\epsilon^2} \log \frac{1}{\epsilon})}$.

5 Conclusion

We studied the minimum liar's dominating set problem (MLDS) on unit disk graphs, showed that the MLDS problem is NP-complete and proposed a PTAS.

References

1. Alimadadi, A., Chellali, M., Mojdeh, D.A.: Liar's dominating sets in graphs. *Discret. Appl. Math.* **211**, 204–210 (2016)
2. Biedl, T., Kant, G.: A better heuristic for orthogonal graph drawings. *Comput. Geom.* **9**(3), 159–180 (1998)
3. Garey, M.R., Johnson, D.S.: *Computers and Intractability: A Guide to the Theory of NP-completeness*. Freeman, Dallas (1979)
4. Hopcroft, J., Tarjan, R.: Efficient planarity testing. *J. ACM (JACM)* **21**(4), 549–568 (1974)
5. Itai, A., Papadimitriou, C.H., Szwarcfiter, J.L.: Hamilton paths in grid graphs. *SIAM J. Comput.* **11**(4), 676–686 (1982)
6. Jallu, R.K., Das, G.K.: Liar's domination in 2D. In: Gaur, D., Narayanaswamy, N.S. (eds.) *CALDAM 2017*. LNCS, vol. 10156, pp. 219–229. Springer, Cham (2017). https://doi.org/10.1007/978-3-319-53007-9_20
7. Nieberg, T., Hurink, J.: A PTAS for the minimum dominating set problem in unit disk graphs. In: Erlebach, T., Persinao, G. (eds.) *WAOA 2005*. LNCS, vol. 3879, pp. 296–306. Springer, Heidelberg (2006). https://doi.org/10.1007/11671411_23
8. Panda, B.S., Paul, S.: Connected liar's domination in graphs: complexity and algorithms. *Discret. Math. Algorithms Appl.* **5**(04), 1350024 (2013)
9. Panda, B.S., Paul, S.: Liar's domination in graphs: complexity and algorithm. *Discret. Appl. Math.* **161**(7), 1085–1092 (2013)
10. Panda, B.S., Paul, S.: A linear time algorithm for liar's domination problem in proper interval graphs. *Inf. Process. Lett.* **113**(19), 815–822 (2013)
11. Panda, B.S., Paul, S.: Hardness results and approximation algorithm for total liar's domination in graphs. *J. Comb. Optim.* **27**(4), 643–662 (2014)
12. Panda, B.S., Paul, S., Pradhan, D.: Hardness results, approximation and exact algorithms for liar's domination problem in graphs. *Theor. Comput. Sci.* **573**, 26–42 (2015)
13. Roden, M.L., Slater, P.J.: Liar's domination in graphs. *Discret. Math.* **309**(19), 5884–5890 (2009)
14. Slater, P.J.: Liar's domination. *Networks* **54**(2), 70–74 (2009)
15. Sterling, C.: *Liar's Domination in Grid Graphs*. Ph.D. thesis, East Tennessee State University (2012)
16. Valiant, L.G.: Universality considerations in VLSI circuits. *IEEE Trans. Comput.* **100**(2), 135–140 (1981)